11/4/2026

Goal: Prove that there are constants $C_1, C_2 > 0$ s.t.

$$C_1 \frac{x}{\ln x} < \pi(x) < C_2 \frac{x}{\ln x}, \quad x \geq 2.$$

Last time:

Candidate: $\binom{2N}{N} = \frac{(2N)!}{N! \cdot N!} \in \mathbb{Z}$.

1. $p \mid \binom{2N}{N} \Rightarrow p \leq 2N$.

2. Size: $2^N \leq \frac{4^N}{2N} \leq \binom{2N}{N} < 4^N, \quad N \geq 1$

3. $e_p\left(\binom{2N}{N}\right) \leq \frac{\ln(2N)}{\ln p}, \quad N \geq 1.$

$$\binom{2N}{N} = \prod_{p \le 2N} p^{e_p\left(\binom{2N}{N}\right)}$$

$$\Rightarrow \underbrace{\ln \binom{2N}{N}}_{\ge 2^N} = \sum_{p \le 2N} \underbrace{e_p\left(\binom{2N}{N}\right)}_{\le \frac{\ln(2N)}{\ln p}} \ln p \le \ln(2N) \underbrace{\sum_{p \le 2N} 1}_{\pi(2N)}$$

$$\Rightarrow N \ln 2 \le \ln(2N) \cdot \pi(2N)$$

$$\Rightarrow \pi(2N) \ge \ln 2 \cdot \frac{N}{\ln(2N)} = \boxed{\frac{\ln 2}{2}} \cdot \frac{2N}{\ln(2N)}.$$

So

$$\pi(N) \ge \underbrace{\boxed{\frac{\ln 2}{2}}}_{= 0.34\ldots} \cdot \frac{N}{\ln N}, \quad N \ge 2 \text{ is even.}$$

But when $N \ge 3$ is odd,

$$\pi(N) \ge \pi(\underbrace{N-1}_{\text{even}}) \ge \frac{\ln 2}{2} \cdot \frac{N-1}{\ln(N-1)}$$

$$= \frac{\ln 2}{2} \cdot \frac{N}{\ln N} \cdot \underbrace{\boxed{\frac{\ln N}{N} \cdot \frac{N-1}{\ln(N-1)}}}_{\ge \text{constant}?}.$$

$$\frac{\ln N}{N} \cdot \frac{N-1}{\ln(N-1)} = \frac{N-1}{N} \cdot \boxed{\frac{\ln N}{\ln(N-1)}}$$

$$> 1$$

$$> \frac{N-1}{N} = 1 - \frac{1}{N} \quad \nearrow$$

$N \geq 3$

(1) $\geq \frac{3-1}{3} = \frac{2}{3}$

$N \geq 5$

(2) $\geq \frac{5-1}{5} = \frac{4}{5}$.

So

$$\pi(N) > \frac{\ln 2}{2} \cdot \frac{4}{5} \cdot \frac{N}{\ln N} = \boxed{\frac{2\ln 2}{5}} \cdot \frac{N}{\ln N}, \quad N \geq 5 \text{ odd}$$

$$= 0.277\ldots$$

$$> \frac{1}{4}$$

When $N = 3$, $\pi(3) = 2 > \frac{2\ln 2}{5} \cdot \frac{3}{\ln 3}$

Hence, $\pi(N) > \frac{1}{4} \cdot \frac{N}{\ln N}$, $N \geq 2$.

Upper bound.

Look at

$$\binom{2n+1}{n} = \frac{(2n+1)!}{n!\,(n+1)!}$$

$$= \frac{(n+2)\cdots(2n+1)}{n!} \in \mathbb{Z}.$$

Observation: If $n+2 \le p \le 2n+1$, then

$$p \mid (n+2)\cdots(2n+1), \quad p \nmid n!$$

$$\Rightarrow p \mid \binom{2n+1}{n}$$

$$\Rightarrow \prod_{n+2 \le p \le 2n+1} p \mid \binom{2n+1}{n}.$$

$$\Rightarrow \prod_{n+2 \le p \le 2n+1} p \le \binom{2n+1}{n}.$$

$$\Rightarrow \sum_{n+2 \le p \le 2n+1} \ln p \le \ln \binom{2n+1}{n}.$$

$$(1+1)^{2n+1} = 2^{2n+1} = \binom{2n+1}{0} + \binom{2n+1}{1} + \cdots +$$

$$\binom{2n+1}{n} + \binom{2n+1}{n+1} + \cdots + \binom{2n+1}{2n+1}$$

$$= \quad =$$

$$\ge 2\binom{2n+1}{n}$$

$$\Rightarrow \binom{2n+1}{n} \le \frac{2^{2n+1}}{2} = 2^{2n} = 4^n.$$

So

$$\sum_{n+2 \le p \le 2n+1} \ln p \le \ln \binom{2n+1}{n} \le \ln 2^{2n} = 2n \ln 2.$$

Define

$$\theta(x) = \sum_{p \le x} \ln p .$$

Remember $\pi(x) = \sum_{p \le x} 1$.

$$\sum_{n+2 \le p \le 2n+1} \ln p \le 2n \ln 2 .$$

$$= \sum_{p \le 2n+1} \ln p - \sum_{p \le n+1} \ln p$$

$$= \theta(2n+1) - \theta(n+1)$$

That is,

$$\theta(2n+1) - \theta(n+1) \le 2n \ln 2 , \quad n \ge 1 .$$

We prove by induction that

$$\theta(n) < 2n \ln 2, \quad n \geq 1.$$

1. $n = 1$: $\theta(1) = 0 < 2 \cdot 1 \cdot \ln 2$. ✓

2. $n = 2$: $\theta(2) = \ln 2 < 2 \cdot 2 \ln 2$. ✓

3. Assume $\theta(n) < 2n \ln 2$ for all $n \leq n_0 - 1$, where $n_0 \geq 3$.

① n_0 is even :

$$\theta(n_0) = \theta(n_0 - 1) < 2(n_0 - 1) \ln 2 < 2 n_0 \ln 2. \quad ✓$$

$n_0 \geq 4$ even

$\Rightarrow n_0$ is composite